

Objective

To build a modular inpatient model that:

- Represents how patients enter, move through, and exit inpatient units
 - Captures the impact of bed availability on ED flow, ICU capacity, and surgical throughput
 - Reflects real operational constraints such as boarding, stepdowns, and delayed discharges
 - Remains interpretable and configurable for scenario testing
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Core Concept: Admission vs Bed Availability

A central principle of the model is:

An admission decision does not mean a bed is available.

Patients frequently wait after admission, and this delay creates pressure across the system.

This leads to two key constructs:

1. Bed Request Queue

Patients waiting for a bed in a specific unit (medicine, surgery, ICU, etc.)

2. Boarding Location

Where the patient physically remains while waiting

Examples:

- ED admit waiting in an ED bed → boarded in ED
- Post-op patient waiting in PACU → boarded in perioperative space
- ICU stepdown waiting for a ward bed → boarded in ICU
- Direct admit waiting at home → in a virtual queue (not physically boarding)

This distinction is critical because physically boarded patients consume scarce resources and impact flow.

Inpatient Unit Design

Each inpatient unit (e.g., medicine, surgery, ICU) will be modeled as:

- A finite number of staffed beds
- A queue of patients requesting admission
- A set of rules determining which patient gets the next available bed

Units may also include:

- Overcapacity beds

- Stepdown relationships (e.g., ICU → medicine)
 - Admission constraints (e.g., capability requirements)
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Competing Demand for Beds

Each unit receives requests from multiple sources:

- Emergency Department admissions
- ICU stepdowns
- Post-operative patients
- Direct/outpatient admissions
- Transfers from other units

These groups compete for the same beds, often with conflicting priorities.

The model will simulate this competition explicitly rather than assuming immediate placement.

Bed Allocation Logic

Instead of simple first-come-first-served, the approach is to use **unit-specific prioritization rules**, considering:

- Source of the patient (ED, ICU, surgery, direct)
- Time spent waiting
- Whether the patient is physically blocking another resource
- Clinical urgency
- Operational priorities (e.g., maintaining surgical flow)

This allows each unit to behave in a way that reflects real hospital decision-making.

Physical vs Virtual Pressure

A key concept in the model is distinguishing between:

- **Physical blockers** (patients occupying beds while waiting)
- **Virtual waiters** (patients not yet consuming hospital space)

Physical blockers (ED boarders, ICU stepdowns, PACU holds) will typically exert greater pressure on bed allocation decisions because they are already impacting capacity.

Length of Stay Modeling

Length of stay will be modeled as a function of:

- Patient characteristics (e.g., diagnosis, age)
- Unit type
- Random variation

In addition, the model will include **delayed discharge effects**, such as patients who remain in hospital due to lack of community placement (e.g., long-term care or social factors). These patients occupy beds longer than clinically required and are a major driver of capacity constraints.

Inter-Unit Flow

Patients may move between units during their stay, such as:

- ICU → medicine (stepdown)
- Medicine → ICU (deterioration)
- Surgery → medicine

This will be modeled using probabilistic transitions to reflect real clinical pathways.

Surgical and Outpatient Integration

Surgical and outpatient activity will be integrated into the inpatient model because:

- Surgical patients consume space (OR, PACU, short-stay beds)
- Some require inpatient admission after procedures
- Lack of downstream beds can cause post-operative boarding and disrupt surgical schedules

This ensures that inpatient capacity constraints are linked to surgical throughput.

Flow Patterns and Timing

Inpatient flow is not constant throughout the day or week. It follows patterns such as:

- Morning discharge peaks
- Reduced weekend discharges
- Time-dependent admission activity

Rather than relying on large datasets, the model will approximate these patterns using simple functions to create realistic flow cycles.

Initial Implementation Approach

The model will be built incrementally to ensure clarity and stability.

Phase 1:

- Define inpatient units with capacity
- Implement bed request queues
- Introduce basic prioritization logic
- Model boarding explicitly

Later Phases:

- Add detailed length-of-stay modeling
 - Introduce inter-unit transitions
 - Incorporate time-of-day flow patterns
 - Refine prioritization and capacity rules
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Overall Intent

The goal is to move beyond isolated ED modeling and represent the hospital as a connected system, where:

- Downstream constraints (inpatient beds) drive upstream performance (ED wait times, ICU availability, surgical flow)
- Boarding is explicitly modeled as a system-level issue
- Operational decisions around bed allocation can be tested and understood

This should allow for more realistic scenario testing and better insight into how capacity changes or policy adjustments affect overall hospital performance.